

Hospital Performance & Patient Outcomes Analytics

A Healthcare Data Analytics Dashboard for Hospital Utilization, Patient Outcomes, and Patient Flow

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Field: Epidemiology & Healthcare Data Analytics

Tools: Python | Pandas | Power BI | DAX | GitHub

Project Type: Healthcare Analytics Portfolio Project

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Executive Summary

Hospital performance analytics plays an important role in understanding patterns of healthcare utilization, patient outcomes, and operational activity. This project developed an interactive healthcare analytics dashboard to examine hospital admissions, length of stay, mortality, readmissions, long-stay admissions, and patient transfers.

The analytical workflow combined Python-based data preparation and feature engineering with Microsoft Power BI for interactive visualization and dashboard development.

The analysis focuses on five major dimensions of hospital performance:

- Hospital utilization
- Length of stay
- Patient outcomes
- Patient flow
- Overall performance patterns

Key indicators developed for the dashboard include total patients, total admissions, average and median length of stay, in-hospital mortality rate, 30-day readmission rate, long-stay rate, and patient transfer activity.

The resulting Power BI dashboard provides an interactive environment for exploring differences across admission types, age groups, and gender. The project is designed as a descriptive healthcare analytics exercise and should not be interpreted as evidence of causal relationships.

Background

Hospitals generate large volumes of data through patient registration, admissions, clinical encounters, transfers, discharge processes, and follow-up activities. When appropriately processed and analyzed, these data can provide valuable information about how healthcare services are utilized and how patients experience hospital care.

Hospital utilization is commonly reflected through indicators such as the number of admissions, length of stay, and patterns of patient movement across hospital units. These indicators can help describe the demand placed on healthcare services and identify areas that may require further operational investigation.

Patient outcomes provide another important dimension of healthcare performance. In-hospital mortality and readmission are particularly relevant because they provide information about patient outcomes following hospitalization. However, these indicators are influenced by multiple factors, including patient characteristics, clinical severity, admission type, and healthcare processes.

Length of stay is also an important hospital performance indicator. Very short stays may reflect rapid treatment or discharge, whereas prolonged stays may be associated with complex care needs, complications, discharge delays, or other operational factors. Therefore, length of stay should be interpreted within the clinical and operational context of the patient population.

Patient flow represents another important component of hospital operations. Transfers between care units can reflect the complexity of patient care and the movement of patients through different stages of hospitalization. Examining transfer patterns can therefore provide additional insight into hospital activity and resource utilization.

This project integrates these dimensions into a single interactive dashboard to demonstrate how healthcare data analytics can support descriptive assessment of hospital performance and generate questions for further investigation.

Problem Statement

Hospital data are often distributed across multiple datasets and contain a mixture of patient-level, admission-level, and operational information. Without systematic data preparation and visualization, it can be difficult to identify important patterns in healthcare utilization and patient outcomes.

The main problem addressed by this project is:

How can hospital admission and operational data be transformed into meaningful healthcare indicators and interactive visualizations that support the assessment of hospital utilization, patient outcomes, and patient flow?

❖ Aim

To develop an interactive healthcare analytics dashboard that describes hospital utilization, patient outcomes, and patient flow using cleaned hospital admission and transfer data.

Objectives

➤ Specific Objectives

The project aims to:

- Clean and assess the quality of hospital-related datasets.
- Prepare an analytical dataset suitable for healthcare analytics.
- Develop relevant healthcare and operational indicators.

- Analyze patterns of hospital admissions.
- Examine the distribution of length of stay.
- Assess in-hospital mortality patterns.
- Examine 30-day readmission patterns.
- Analyze long-stay admissions.
- Explore patient transfer activity and patient flow.
- Develop an interactive Power BI dashboard for communicating findings.

Key Research Questions

Analytical Questions

The project addresses the following questions:

- How is the hospital being utilized?
- Which admission types account for the largest number of admissions?
- How is length of stay distributed across admissions?
- How does average length of stay vary across patient groups?
- What patterns are observed in in-hospital mortality?
- How common are 30-day readmissions among eligible admissions?
- Which patient groups have higher long-stay rates?
- How frequently are patients transferred between care units?
- Do patient flow patterns differ by admission type?
- What major patterns can be identified across hospital performance indicators?

Data Source

The project uses hospital-related patient, admission, and transfer datasets prepared for analytical purposes.

The main analytical dataset used in Power BI is:

hospital_analytics.csv

An additional operational dataset is:

transfers_clean.csv

The datasets were processed using Python before being imported into Power BI.

Important: Before publishing the final report, we should insert the exact original dataset name, source, version, and citation. We should not invent these details.

Dataset Structure

The analytical dataset contains patient-level and admission-level variables used to examine hospital utilization and outcomes.

Important variables include:

- subject_id
- hadm_id
- gender
- anchor_age
- age_group
- admittance
- dischtime
- admission_type
- admission_type_group
- length_of_stay_days
- los_category
- long_stay
- in_hospital_death
- transfer_count
- readmission_eligible
- readmission_30d

- admission_count
- multiple_admissions

Data Cleaning & Quality Assessment

Data preparation was conducted using Python and Pandas.

The cleaning process included:

- Reviewing dataset structure
- Inspecting variable types
- Identifying missing values
- Checking duplicate records
- Standardizing variables
- Converting date and time variables
- Checking logical consistency
- Preparing variables for analysis
- Creating an analytical dataset for Power BI

The objective of the cleaning stage was to improve data consistency and ensure that the resulting dataset was appropriate for descriptive healthcare analysis.

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Healthcare Indicators

Key Healthcare Indicators

Indicator	Definition
Total Patients	Number of unique patients
Total Admissions	Number of unique hospital admissions
Average LOS	Mean length of hospital stay
Median LOS	Median length of hospital stays
Mortality Rate	In-hospital deaths divided by admissions
30-Day Readmission Rate	Eligible admissions followed by readmission within 30 days
Long-Stay Rate	Long-stay admissions divided by total admissions
Average Transfers	Average transfer activity per admission

Methodology

Analytical Methodology

The project followed a structured healthcare data analytics workflow:

Raw Data



Data Understanding



Data Cleaning



Quality Assessment



Feature Engineering



Healthcare Indicators



Exploratory Analysis



Power BI Data Model



Dashboard Development



Performance Interpretation

Power BI Dashboard

Dashboard Development

The cleaned analytical dataset was imported into Power BI to develop an interactive dashboard.

The dashboard consists of six pages:

Page 1 — Executive Overview

Provides a high-level summary of:

- Total patients
- Total admissions
- Average LOS
- Mortality
- Readmission

Page 2 — Hospital Utilization

Examines:

- Admission patterns
- Length of stay
- Long-stay admissions
- Age-related utilization patterns

Page 3 — Patient Outcomes

Focuses on:

- In-hospital mortality
- 30-day readmission
- Mortality by age group
- Mortality by admission type
- Readmission patterns

Page 4 — Patient Flow

Examines:

- Patient transfers

- Transfer frequency
- Care-unit movement
- Multiple admissions
- Relationship between transfers and LOS

Page 5 — Performance Analysis

Combines major indicators to examine:

- Mortality
- Readmission
- Length of stay
- Long-stay admissions
- Admission patterns

Results

The results section will be completed using the actual values from the final Power BI dashboard.

We should **not insert estimated or invented values** at this stage.

The final results will report:

- Total number of patients
- Total admissions
- Average LOS

- Median LOS
- Mortality rate
- 30-day readmission rate
- Long-stay rate
- Transfer activity
- Differences across age groups
- Differences across admission types

This approach ensures that the written report remains fully consistent with the final analytical dataset and dashboard.

Key Findings

After extracting the actual dashboard results, findings will be organized into five domains:

Hospital Utilization

Patterns in admissions and length of stay.

Patient Outcomes

Patterns in mortality and readmission.

Patient Characteristics

Differences across age groups and gender.

Patient Flow

Patterns in transfers and multiple admissions.

Performance

Differences across admission types and patient groups.

Discussion

The dashboard provides a multidimensional view of hospital activity by combining utilization, patient outcomes, and patient flow indicators.

Analyzing these dimensions together is important because no single indicator provides a complete picture of hospital performance. For example, a high length of stay may reflect patient complexity rather than poor performance, while a high mortality rate may reflect a population with greater clinical severity.

Similarly, differences in readmission rates should be interpreted alongside patient characteristics, admission type, length of stay, and other relevant clinical and operational factors.

The dashboard therefore serves primarily as a **descriptive and exploratory analytical tool**. It can help identify patterns that warrant deeper investigation using additional clinical, socioeconomic, or operational data.

Limitations

Several limitations should be considered:

1. The analysis is primarily descriptive.
2. Observed associations should not be interpreted as causal relationships.
3. Available variables may not capture all clinical factors influencing outcomes.
4. Mortality may be influenced by disease severity and case mix.
5. Readmission patterns may be influenced by factors outside the hospital.
6. Length of stay may be affected by clinical and operational factors.
7. Transfer patterns depend on the completeness of transfer records.
8. Some indicators may require additional clinical or administrative data for more comprehensive interpretation.

Recommendations

Based on the analytical framework, future analysis could focus on:

1. Investigating factors associated with prolonged length of stay.
2. Conducting risk-adjusted mortality analysis.
3. Investigating predictors of 30-day readmission.
4. Examining transfer patterns at the care-unit level.
5. Incorporating diagnosis and clinical severity variables.
6. Monitoring hospital performance indicators over time.
7. Developing predictive models for readmission and prolonged hospitalization.
8. Integrating additional operational indicators such as bed occupancy and resource utilization where appropriate.

Conclusion

This project demonstrates how healthcare data can be transformed into actionable descriptive insights through a combination of Python-based data preparation and Power BI visualization.

The resulting dashboard provides an integrated view of hospital utilization, length of stay, patient outcomes, and patient flow.

By combining multiple indicators rather than relying on a single measure of performance, the dashboard provides a more comprehensive framework for exploring hospital activity and identifying areas for further investigation.

The project also demonstrates the potential of healthcare analytics to support evidence-informed decision-making, monitoring, and hypothesis generation.

